9/30/24

Recall task: Solve LSP

In preparation: ips, orthogonal matrices
(& will continue)

To solve LSP: (options)
- projections in ips
- normal equations
- pseudoinverse (Moore-Penrose)
- Schur's Theorem (SVD)
- QR decompositions * ← will do this.

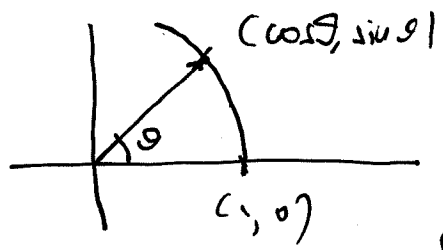
More on orthogonal matrices: $Q^T Q = I$ $(Q^T = Q^{-1})$

Two basic constructions of orthogonal matrices

(1) Rotators

In $\mathbb{R}^2$: Consider $Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

$\theta \in [0, 2\pi)$




$Q\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$

$Q$ rotates $e_1$ (& $e_2$) by $\theta$ counterclockwise.

To generalize: Seek an orthogonal $Q$ so that $Q^T x = Q^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y \\ 0 \end{bmatrix}$, $y \in \mathbb{R}$.

(introduce a zero in the second entry)

Use $Q = \begin{bmatrix} \cos\vartheta & -\sin\vartheta \\ \sin\vartheta & \cos\vartheta \end{bmatrix}$ (orthogonal!)

$$Q^T x = \begin{bmatrix} x_1 \cos\vartheta + x_2 \sin\vartheta \\ -x_1 \sin\vartheta + x_2 \cos\vartheta \end{bmatrix} \overset{\text{want}}{=} \begin{bmatrix} y \\ 0 \end{bmatrix}$$

i.e. $\vartheta$ should be chosen so that

$$-x_1 \sin\vartheta + x_2 \cos\vartheta = 0 \qquad (*)$$

$$\Rightarrow \quad x_1 \sin\vartheta = x_2 \cos\vartheta$$

$$\Rightarrow \quad \tan\vartheta = x_2 / x_1$$

$$\Rightarrow \quad \vartheta = \arctan\left(\frac{x_2}{x_1}\right)$$

We can actually find $\vartheta$ by noticing that $(*)$ is satisfied if

$$\left.\begin{aligned} x_1 &= \cos\vartheta \\ x_2 &= \sin\vartheta \end{aligned}\right\} \Rightarrow x_1^2 + x_2^2 = 1$$

So we can take $\vartheta$ so that

$$\cos\vartheta = \frac{x_1}{\sqrt{x_1^2 + x_2^2}}, \quad \sin\vartheta = \frac{x_2}{\sqrt{x_1^2 + x_2^2}}$$

$$= \frac{x_1}{\|x\|_2} \qquad = \frac{x_2}{\|x\|_2}$$

Then $Q^T x = \begin{bmatrix} \frac{x_1}{\|x\|_2} & -\frac{x_2}{\|x\|_2} \\ \frac{x_2}{\|x\|_2} & \frac{x_1}{\|x_1\|_2} \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$= \cdots = \begin{bmatrix} \frac{x_1^2 + x_2^2}{\|x\|_2} \\ 0 \end{bmatrix} = \begin{bmatrix} y \\ 0 \end{bmatrix}, \quad y = \sqrt{x_1^2 + x_2^2}$$

For matrix $A \in M_2(\mathbb{R})$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$Q^T \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} = \begin{bmatrix} r_{11} \\ 0 \end{bmatrix}$$

appropr. chosen as above

$$Q^T \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix} = \begin{bmatrix} r_{12} \\ r_{22} \end{bmatrix}$$

So $\quad Q^T A = \begin{bmatrix} r_{11} & r_{12} \\ 0 & r_{22} \end{bmatrix} = R$ = upper triangular

Since $Q^T = Q^{-1}$,

$$A = QR \leftarrow \text{by QR-decomposition}$$

Illustrate use in solving systems:

$$Ax = b$$

Factor $A = QR$, $Q$ orthog., $R$ upper tri.

$$QRx = b \Rightarrow Rx = Q^T b = c$$

Solve $Rx = c$ by back substitution

<u>Example</u> Solve $\begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

Want $Q^T \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} * \\ 0 \end{bmatrix}$

$$Q = \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$$

$$Q^T \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \cdots = \begin{bmatrix} \sqrt{5} \\ 0 \end{bmatrix}$$

$$c = Q^T b = Q^T \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 4 \\ -3 \end{bmatrix}$$

Solve $\underbrace{\begin{bmatrix} \sqrt{5} & 9 \\ 0 & 2 \end{bmatrix}}_{R} \underbrace{\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}}_{} = \underbrace{\frac{1}{\sqrt{5}} \begin{bmatrix} 4 \\ -3 \end{bmatrix}}_{c}$

$$\Rightarrow z_2 = \frac{-3}{2\sqrt{5}}, \quad z_1 = \cdots = -\frac{1}{5}\left(\frac{4}{\sqrt{5}} + \frac{27}{2\sqrt{5}}\right)$$

## Generalize

Consider $Q = \begin{bmatrix} 1 & & & & & & & 0 \\ & \ddots & & & & & & \\ & & 1 & & & & & \\ & & & c & & & -s & \\ & & & & 1 & & & \\ & & & & & \ddots & & \\ & & & s & & & c & \\ 0 & & & & & & & \ddots \\ & & & & & & & & 1 \end{bmatrix}$ ← $i$th, ← $j$-th

$= Q_{ij}$

e.g. $\begin{bmatrix} c & 0 & -s & 0 \\ 0 & 1 & 0 & 0 \\ s & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = Q_{13}$

i.e. We can embedd $\begin{bmatrix} c & -s \\ s & c \end{bmatrix}$ as a principal submatrix of the identity $I$ & take $c = \cos\theta$, $s = \sin\theta$.

The resulting matrix $Q$ is orthogonal (prove please) & it rotates vector $x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ in the $(i,j)$-plane by $\theta$ counter clockwise, when $\alpha = \{i,j\}$ define the principal submatrix $I[\alpha]$ used in creating $Q$.

This can be used for a general QR factorization.

So

<u>Theorem</u> Let $A \in M_n(\mathbb{R})$. Then there exists orthogonal $Q \in M_n(\mathbb{R})$ & upper triangular $R \in M_n(\mathbb{R})$ such that

$$A = QR$$

<u>Proof</u> (TBC) Sketch/more rigorous proof shortly. 10/2/24

Given $A$, we'll construct $Q$ as a product of rotators:

$$\exists Q_{21} \text{ orth}: \quad Q_{21}^T A = \begin{bmatrix} * & \\ 0 & \\ * & * \\ \vdots & \\ * & \end{bmatrix} \quad \text{(like in the 2x2 example)}$$

$$\exists Q_{31} \text{ orth}: \quad Q_{31}^T (Q_{21}^T A) = \begin{bmatrix} * & \\ 0 & \\ 0 & \\ * & * \\ \vdots & \\ * & \end{bmatrix}$$

$$\vdots$$

$$= \begin{bmatrix} * & * \\ 0 & \\ \vdots & \boxed{\tilde{A}_1} \\ 0 & \end{bmatrix}$$

$\hookrightarrow (n-1)\times(n-1)$

Continue working with $\tilde{A}_1$ & zero-out the entries below final diagonal entry in its first column. So

$$\underbrace{Q_{n\,n-1}^T Q_{n\,n-2}^T \cdots Q_{n1}^T \cdots Q_{21}^T}_{= Q^T} A = R = \text{upper triangular}$$

10/2/24

where $Q$ is orthogonal (prove that the set of $n \times n$ orthogonal matrices is a multiplicative group)

Thus $Q^T A = R \Rightarrow A = QR$ ■

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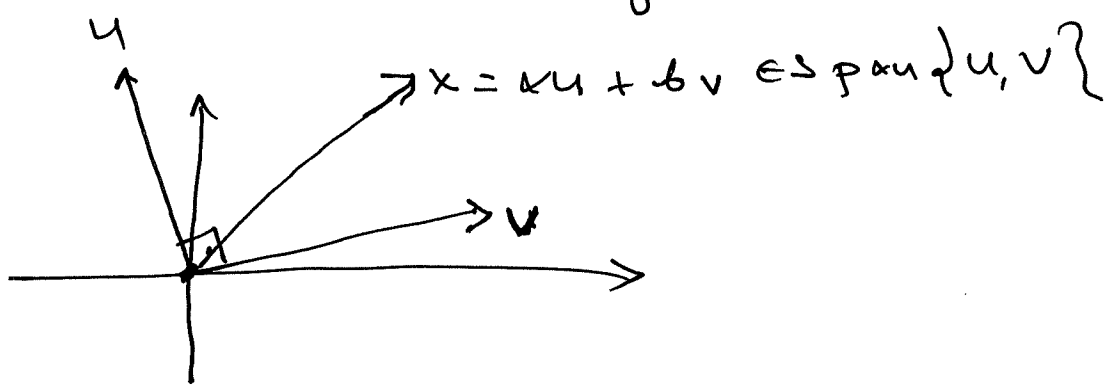
(2) Reflectors : another type of orthogonal matrix.




Objective : Find orthogonal $Q$ such that

$$Qx = -\alpha u + \beta v$$

i.e. $Qx$ is the reflection of $x$ through the span$\{v\}$

$Q = ??$

The type of $Q$ that achieves that is described next.

Given $v \neq 0$, $u$ in $\mathbb{R}^n$, $\|u\|_2 = 1$, $u \perp v$

$\langle u, v \rangle = v^T u = 0$

define $P = uu^T \in M_n(\mathbb{R})$. Notice:

$$Pu = uu^T u = u\|u\|_2^2 = u$$

$$Pv = uu^T v = u \cdot 0 = 0$$

Next define

$$Q = I - 2P$$

so that

$$Qu = u - 2Pu = u - 2u = -u$$

$$Qv = v - 2Pv = v - 0 = v$$

<u>Theorem 1</u> Let $u \in \mathbb{R}^n$, $\|u\|_2 = 1$. Define $P = uu^T \in M_n(\mathbb{R})$. Then

(a) $Pu = u$

(b) $Pv = 0$ if $u^T v = 0$

(c) $P^2 = P$ ($P$ is "idempotent" or "projector")

(d) $P^T = P$

<u>Proof</u> (a) & (b) see above. Clearly

$$P^2 = uu^T uu^T = uu^T = P$$

$$P^T = (uu^T)^T = (u^T)^T u^T = uu^T = P \quad \blacksquare$$

<u>Theorem</u>[2] Let $u \in \mathbb{R}^n$, $\|u\|_2 = 1$, $Q = I - 2uu^T$. Then

(a) $Qu = -u$

(b) $Qv = v$ if $u^T v = 0$

(c) $Q^T = Q$ (symmetric)

(d) $Q^T = Q^{-1}$ (orthogonal)

(e) $Q^{-1} = Q$ (involution)

<u>Proof</u> (a) & (b) see above. (c) is easy.

Also

$$\begin{aligned} Q^T Q &= (I - 2uu^T)^T (I - 2uu^T) \\ &= (I - 2uu^T)(I - 2uu^T) \\ &= I - 2uu^T - 2uu^T + 4uu^T uu^T \\ &= I, \end{aligned}$$

so (d) holds, and of course (e). ∎

<u>Terminology</u> By the above theorems, multiplication by $Q$ of a vector in $\text{span}\{u, v\}$ is reflected across $\text{span}\{v\}$.

$Q$ is called a <u>reflector</u> or a <u>Householder transformation</u>

<u>Theorem</u> (3.2.31) Let $x, y \in \mathbb{R}^n$, $x \neq y$, $\|x\|_2 = \|y\|_2$. There exists unique reflector $Q$ such that $Qx = y$.

<u>Proof</u>

Find $u$ such that $(I - \gamma u u^T)x = y$ where $\gamma = \frac{2}{\|u\|_2^2}$. To do that

$$u = x - y, \quad \gamma = \frac{2}{\|u\|_2^2}, \quad Q = I - \gamma u u^T$$

Then

$$x = \frac{1}{2}(x+y) + \frac{1}{2}(x-y)$$

so

$$Qu = Q(x-y) = Iu - \gamma u u^T u$$

$$= u - \frac{2}{\cancel{u^T u}} u \cancel{u^T u}$$

$$= -u$$

i.e. $\boxed{Q(x-y) = y - x}$ (#)

Also

$$\langle x+y, x-y \rangle = \langle x, x \rangle - \cancel{\langle x, y \rangle} + \cancel{\langle y, x \rangle} - \langle y, y \rangle$$

$$= \|x\|_2^2 - \|y\|_2^2 = 0.$$

Thus $(x+y) \perp (x-y)$, i.e.

$$Q(x+y) = (x+y) - \frac{2}{\|u\|_2^2} uu^T(x+y)$$

$u = x-y$

$= \dots = x+y$, i.e.

$$\boxed{Q(x+y) = x+y} \quad (\#\#)$$

So if we let $v = x+y$, we have

$$Qv = Q(x+y) = x+y = v$$

$$\Rightarrow \begin{cases} Qx - Qy = x-y \\ Qx + Qy = x+y \end{cases}$$

$$\Rightarrow 2Qx = 2y \Rightarrow Qx = y \checkmark$$

<u>Uniqueness</u> Assume $\hat{Q}x = y$ for some reflector $\hat{Q}$. Prove $\hat{Q} = Q$ (Exercise)

$$(Qx = y = \hat{Q}x \Rightarrow \dots)$$

<u>Corollary</u>

$\forall x \in \mathbb{R}^n$, $\exists!$ reflector $Q$

such that

$$Qx = \begin{bmatrix} * \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

where $* = \pm \|x\|_2$

We'll use this to factor $A = QR$

TBC.

10/4/24

# Revisit QR-decomposition via reflectors

**Theorem** (3.2.21 again!)

Every $A \in M_n(\mathbb{R})$ can be factored into $A = QR$, $Q$ = orthogonal, $R$ = upper triangular.

**Proof** Induction on $n$

$n=1$ $\quad A = [a_{11}]$, $Q = [1]$, $R = [a_{11}]$

$n \geq 2$ $\quad$ Assume theorem hold for $(n-1)\times(n-1)$ matrices

$A = [a_{ij}]$

Let $Q_1 \in M_n(\mathbb{R})$ be a reflector so that

$$Q_1^T \begin{bmatrix} a_{11} \\ \vdots \\ a_{n1} \end{bmatrix} = \begin{bmatrix} -\tau_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

and thus

$$Q_1^T A = Q_1^T A = \left[\begin{array}{c|ccc} -\tau_1 & \hat{a}_{12} & \cdots & \hat{a}_{1n} \\ \hline 0 & & & \\ \vdots & & \hat{A}_2 = \hat{Q}_2 \hat{R}_2 & \\ 0 & & & \end{array}\right]$$

hypothesis

Let $$\tilde{Q}_2 = \left[\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \hat{Q}_2 & \\ 0 & & & \end{array}\right]$$ : also orthogonal & reflector

Then if we consider the fact that

$$\underbrace{\tilde{Q}_2 Q_1^T}_{Q^{-1}} A = \underbrace{\left[\begin{array}{c|ccc} -\tau_1 & \hat{a}_{12} & \cdots & \hat{a}_{1n} \\ \hline 0 & & & \\ \vdots & & \hat{R}_2 & \\ 0 & & & \end{array}\right]}_{R}$$

$$\Rightarrow \quad Q^{-1} A = R \quad \Rightarrow \quad A = QR \quad \blacksquare$$

## Third proof of QR-Factorization

The one to remember!

**Theorem** (3.2.45) Let $A \in M_n(\mathbb{R})$ be nonsingular. Then there exist unique orthogonal $Q$ and upper triangular $R$ with positive diagonal entries, such that $A = QR$.

**Proof** Existence of $A = \hat{Q}\hat{R}$, $\hat{Q}$ = orthogonal, $\hat{R}$ = upper triangular has been accomplished via rotations or reflections above.

Since $A$ is nonsingular, let $\hat{R} = [\hat{r}_{ij}]$ with $\hat{r}_{ii} \neq 0$, $i = 1, 2, \ldots, n$. Define the diagonal matrix

$$D = [d_{ii}] \in M_n(\mathbb{R})$$

such that

$$d_{ii} = \begin{cases} 1 & \text{if } \hat{r}_{ii} > 0 \\ -1 & \text{if } \hat{r}_{ii} < 0 \end{cases}, \quad i = 1, 2, \ldots, n$$

Let $Q := \hat{Q} D^{-1}$, $R := D\hat{R}$

$(D = D^T = D^{-1})$

So $Q$ is orthogonal, $R$ is upper triangular with positive diags

$$\left( \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right)$$

Finally, $A = \hat{Q}\hat{R} = QDD^{-1}R = QR$

as required/claimed.

Now we can prove uniqueness:

Say $\boxed{A = Q_1 R_1 = Q_2 R_2}$

$Q_1, Q_2$ = orthogonal

$R_1, R_2$ = upper tr. with pos. diags

Then

$$A^T A = R_1^T \underbrace{Q_1^T Q_1} R_1 = R_1^T R_1$$

& also $A^T A = R_2^T R_2$

But $A^T A$ is positive definite & $R_1^T R_1$, $R_2^T R_2$ are Cholesky factorizations of $A^T A$. We have proven that such factorizations are unique!

i.e. $\boxed{R_1 = R_2}$

So $A = \underbrace{Q_1 R_1 = Q_2 R_1} \Rightarrow \boxed{Q_1 = Q_2}$ ∎

<u>Note</u> Hidden in the argument above is that $R_1^T R_1$ is positive definite because $x^T R_1^T R_1 x > 0$ $\forall x \neq 0$ since the diagonal entries of $R_1$ are positive.

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<u>Historical comment</u>: The orthogonality of the columns of $Q$ in the factorization $A = QR$ is the basis of famous result by Wilkinson that the QR-factorization is "numerically stable".

## § 3.3 Solution of LSP
(back to it...)

Recall the context:

- unknown function $y(t)$
- few observations $(t_i, y(t_i))$
- assume something about the function $y$ (e.g linear, quadratic)
- Set-up the problem of finding $y(t)$ (i.e. approximating $y(t)$ at all other $t$'s) as an over-determined system

$$A\,x = b$$

given observations

$n \times m$
$n \gg m$

unknown values of $y(t)$
$x \in \mathbb{R}^m$

<u>Recall</u>: If $Ax=b$ is consistent, the solution $x$ is unique <u>iff</u> $\operatorname{rank} A = m$